

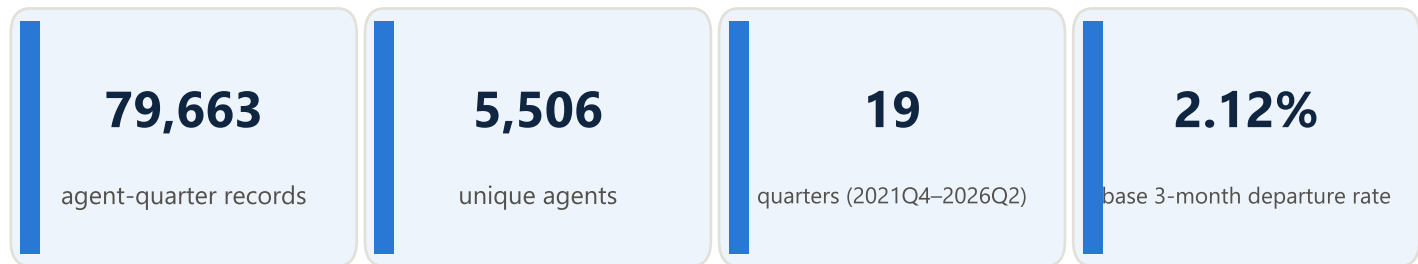
AGENT LIKELIHOOD-TO-LEAVE MODEL

Model Performance & Feature Summary

Rhode Island (RIAR) real estate agents | July 2026

73.7%

AUC (ranking accuracy) — CatBoost, 5-fold time-based validation



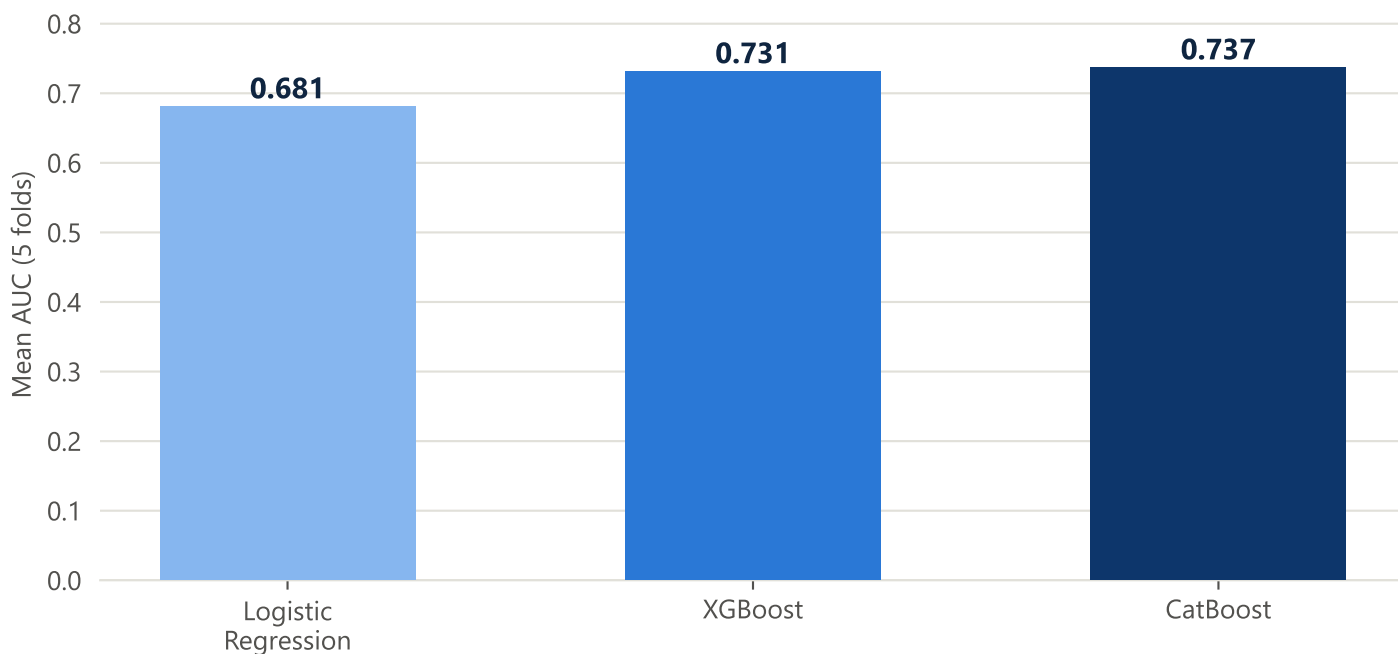
What this model does

Predicts each agent's percent chance of switching brokerages in the next 3 months, for rank-based (top-N) retention outreach. Output: a calibrated percent chance, a risk multiplier vs. the base rate, and a 5-tier risk label (Low → High).

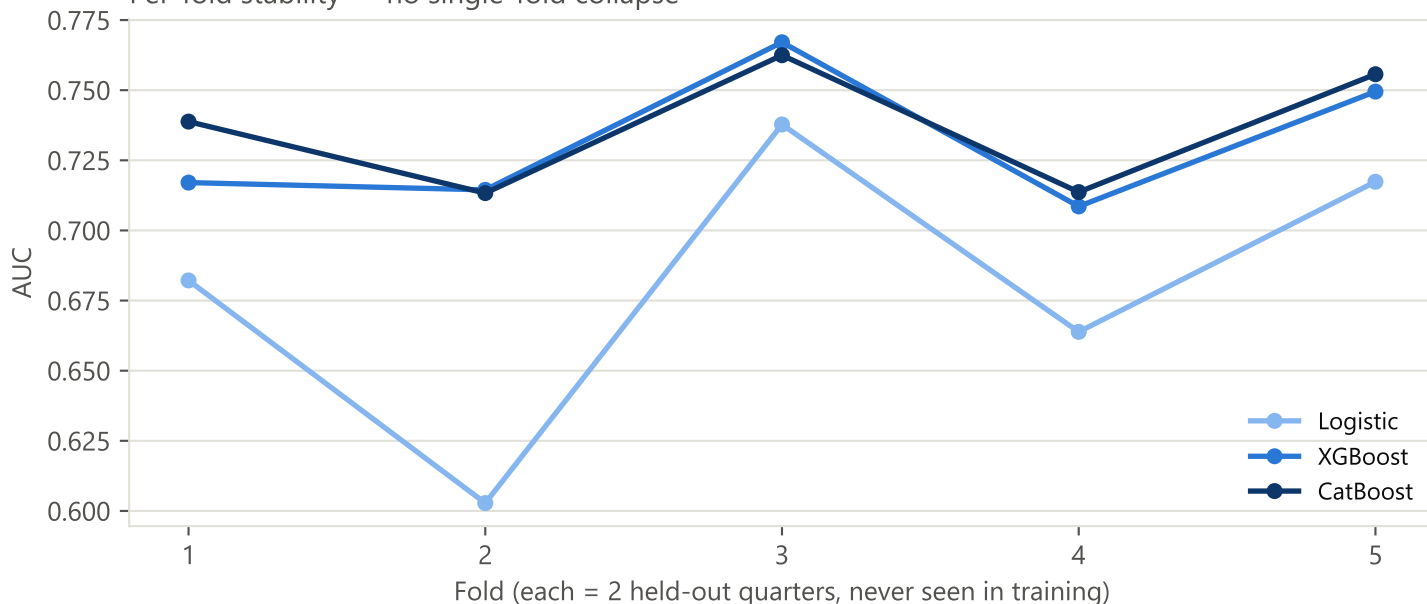
Model Comparison

Three models compared on 5 forward-chaining (time-based) folds

Mean ranking accuracy by model



Per-fold stability — no single-fold collapse

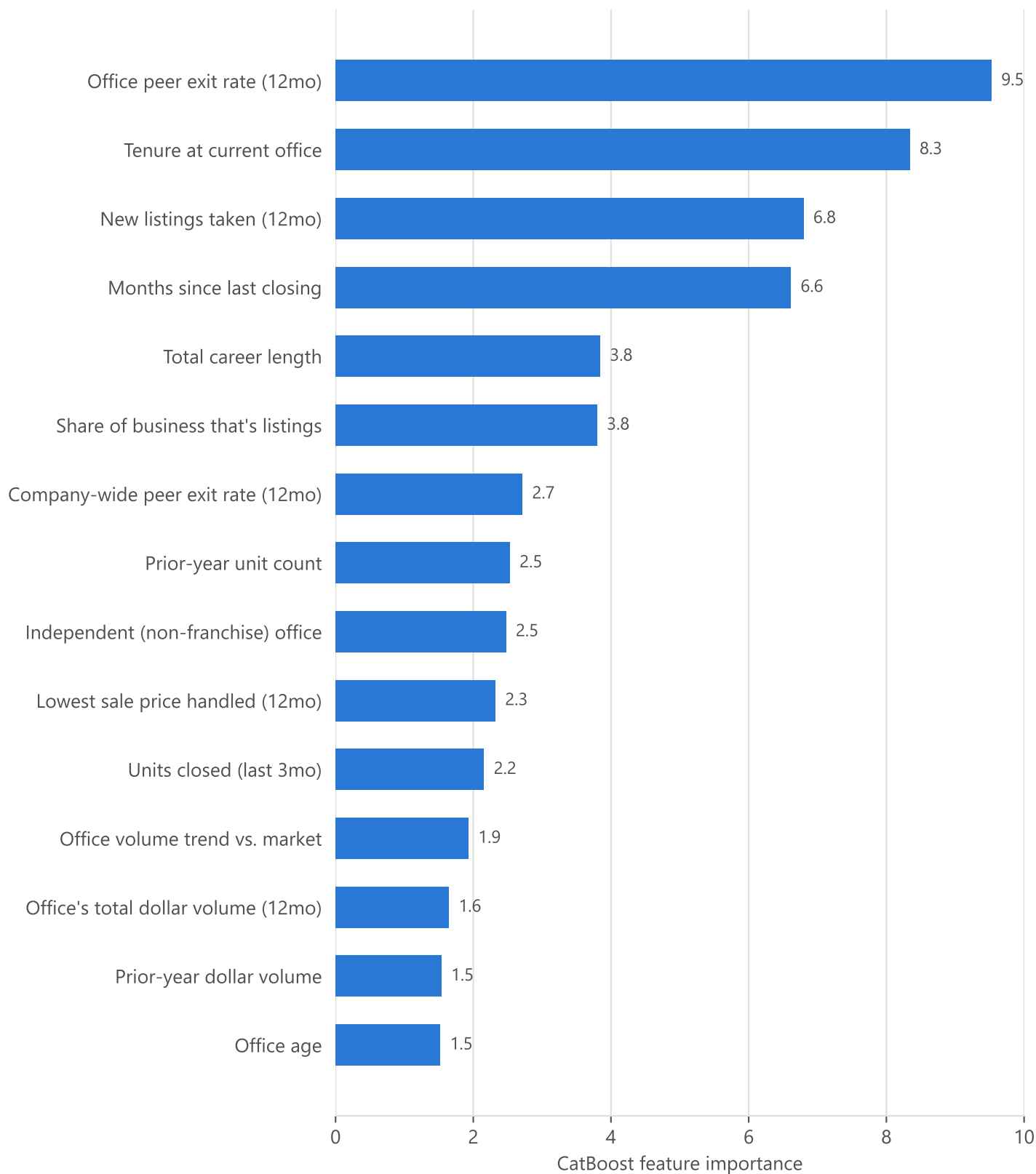


Why CatBoost:

Wins 4 of 5 folds, best mean AUC and PR-AUC, and lift-at-top-decile of 3.7x the base rate (vs. 3.6x for XGBoost). The speed gap (~7s vs ~1.4s per fold) doesn't matter at a quarterly retrain cadence. Logistic regression is kept only as an interpretability gut-check, not for production scoring.

What Drives the Prediction

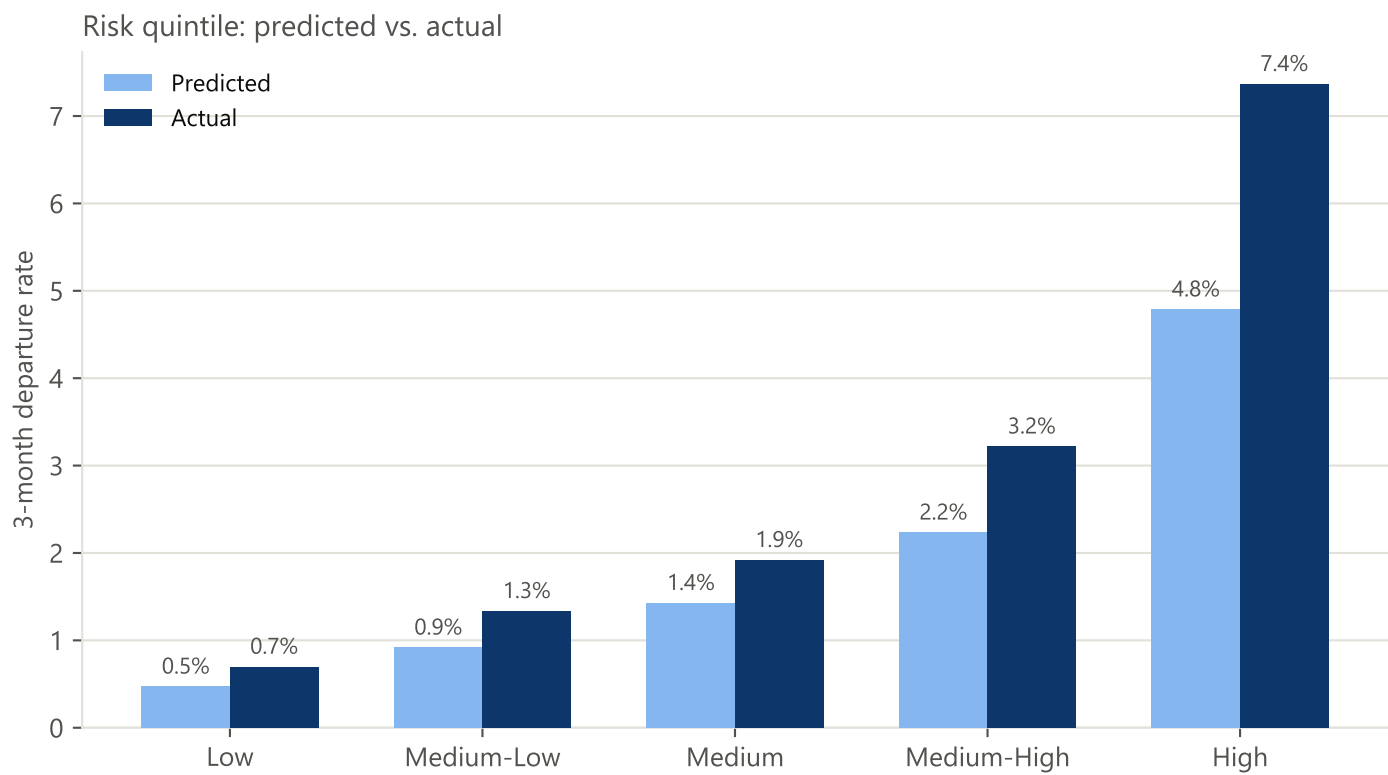
Top 15 features by CatBoost importance (of ~95 total)



New this round: company-level exit rate and office franchise brand both rank in the top 9 — unused signals that turned out to matter once wired in.

Calibration: Do the Percentages Mean What They Say?

Predicted vs. actual departure rate, by risk quintile (out-of-fold, bias-corrected)



Risk output per agent

Quintile	Cutpoint (calibrated %)	Actual rate	Risk multiplier
Low	< 0.72%	0.70%	0.3x base rate
Medium-Low	0.72%–1.14%	1.33%	0.6x base rate
Medium	1.14%–1.79%	1.92%	0.9x base rate
Medium-High	1.79%–2.90%	3.22%	1.5x base rate
High	≥ 2.90%	7.37%	3.5x base rate

Bias correction & honest uncertainty

A backtest found the calibrator over-predicting in recent quarters (traced to a spike in confirmed rebrand/M&A cleanup, not real behavior change). A bias correction of 0.68x is applied to today's output, refreshed every quarterly retrain, plus a $\pm 15\%$ plausible range shown alongside each percentage.

Methodology at a Glance

Key decisions behind the numbers

Prediction horizon

3 months (not 12) — unlocks 94% of rows vs. 78%, matches “who's a flight risk right now.”

Validation

Forward-chaining time-based folds, never random k-fold — agents recur across quarters, so a random split would leak the future into the past.

Label cleanup

335 rows recoded from “left” to “stayed” across 23 confirmed rebrand/M&A/mass-mover office transitions (≥ 5 agents, same origin/destination, same quarter) — corporate events, not voluntary departures.

Features

~95 features: agent, office, company, and market-level activity and trends, plus macro indicators (state unemployment, CPI) and market-relative performance ratios.

Class imbalance

Handled via class-weighting (not synthetic oversampling) — only ~2.1% of rows are positive.

Calibration

Platt scaling with a quarterly bias-correction step (see previous page), not a fixed historical calibration.

Data source

79,663 agent-quarter records, 5,506 unique RIAR (Rhode Island) agents, 19 quarterly snapshots (2021 Q4 – 2026 Q2).